ANSWER KEY & MARKING SCHEME SET 1

Chapter 2: **Acids, Bases and Salts** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** sour	Acids taste sour.
2	**(d)** 7	Neutral solutions have pH 7.
3	**(a)** salt and water	Neutralisation.
4	**(c)** soapy	Bases are soapy to touch.
5	**(d)** acidic	pH below 7 is acidic.
6	**(d)** hydrogen gas	Hydrogen is released.
7	**(a)** acid	Acids turn blue litmus red.
8	**(a)** basic	pH above 7 is basic.
9	**(d)** carbon dioxide	CO_2 is evolved.
10	**(a)** base	Bases turn red litmus blue.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Substance giving H^+ ions in water (1); HCl, H_2SO_4 (1).	1+1
12	Colourless in acid (1); Pink in base (1).	1+1
13	Acid + base (1); gives salt + water (1).	1+1
14	Substance that gives OH^- ions in water / reacts with acids (1); NaOH, KOH (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Lemon juice: blue litmus turns red (1); Soap solution: red litmus turns blue (1); Phenolphthalein: colourless in lemon juice, pink in soap (1).	1+1+1
16	Sour taste (1); Turn blue litmus red (1); React with metals to give hydrogen (1).	1+1+1
17	pH below 5.6 is acid rain (1); It makes river water acidic (1); Survival of aquatic life becomes difficult (1).	1+1+1
	OR: Magnesium chloride and hydrogen (1); $Mg + 2\,HCl \longrightarrow MgCl_2 + H_2$ (1); Hydrogen burns with a pop sound (1).	1+1+1
18	Bitter taste and soapy touch (1); Turn red litmus blue (1); React with acids to give salt and water (1).	1+1+1
	OR: Methanoic acid is injected (1); Rub a mild base like baking soda (1); It neutralises the acid (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Acid sour, base bitter (1); Acid turns blue litmus red, base turns red litmus blue (1); Acid gives H^+, base gives OH^- (1); Acid pH below 7, base pH above 7 (1); Examples: HCl, NaOH (1).	1+1+1+1+1
	OR: $Na_2CO_3 \cdot 10\,H_2O$; prepared from baking soda (1); Uses: glass, soap, hard water (1); $CaSO_4 \cdot \frac{1}{2}\,H_2O$; heat gypsum at 373 K (1); $CaSO_4 \cdot 2\,H_2O \longrightarrow CaSO_4 \cdot \frac{1}{2}\,H_2O + 1.5\,H_2O$ (1); Uses: bone plaster, toys (1).	1+1+1+1+1
20	$Zn + 2\,HCl \longrightarrow ZnCl_2 + H_2$ (1); $NaOH + HCl \longrightarrow NaCl + H_2O$ (1); $Na_2CO_3 + 2\,HCl \longrightarrow 2\,NaCl + H_2O + CO_2$ (1); $CuO + 2\,HCl \longrightarrow CuCl_2 + H_2O$ (1); $NaHCO_3 + HCl \longrightarrow NaCl + H_2O + CO_2$ (1).	1+1+1+1+1
	OR: Fixed number of water molecules in a salt unit (1); Washing soda: 10 (1); Blue vitriol: 5 (1); Gypsum: 2 (1); Plaster of Paris: half (1).	1+1+1+1+1

Mistakes to avoid:

- Pouring water into concentrated acid; always add the acid slowly to water with stirring because the process is highly exothermic.
- Writing that a solution with pH 3 is 'less acidic' than one with pH 5; a lower pH means a **stronger** acid.
- Mixing up the formulae: baking soda is $NaHCO_3$, washing soda is $Na_2CO_3 \cdot 10\,H_2O$, plaster of Paris is $CaSO_4 \cdot \frac{1}{2}\,H_2O$ and bleaching powder is $CaOCl_2$.
- Thinking that all compounds containing hydrogen are acids; glucose and alcohol do not release H^+ ions.
- Writing H_2 as the gas formed when an acid reacts with a carbonate; the gas is CO_2.
- Assuming neutralisation always gives a neutral solution; a strong acid with a weak base gives an acidic salt solution (and vice-versa).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 2: **Acids, Bases and Salts** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** the approximate pH	It shows different colours for different pH.
2	**(c)** hydrochloric acid	Stomach acid is HCl.
3	**(d)** acetic acid	Vinegar is dilute acetic acid.
4	**(b)** HCl	Hydrochloric acid is HCl.
5	**(a)** about 2	Lemon juice is acidic.
6	**(d)** citric acid	Lemon contains citric acid.
7	**(a)** NaOH	Caustic soda is NaOH.
8	**(c)** 7.4	Blood is slightly basic.
9	**(b)** lactic acid	Curd contains lactic acid.
10	**(d)** NaCl	Sodium chloride.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Red in acid (1); Yellow in base (1).	1+1
12	Acetic acid (1); Citric acid (1).	1+1
13	A base that dissolves in water (1); NaOH or KOH (1).	1+1
14	Potential of hydrogen (1); 0 to 14 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Sodium chloride and water (1); $\mathrm{NaOH + HCl \longrightarrow NaCl + H_2O}$ (1); Neutralisation (1).	1+1+1
16	Dilution is exothermic (1); Water on conc. acid may splash (1); Adding acid to water spreads heat safely (1).	1+1+1
17	Lemon, orange (1); Tomato, spinach (1); Both occur naturally in food (1).	1+1+1
	OR: Sodium sulphate, water and carbon dioxide (1); $\mathrm{Na_2CO_3 + H_2SO_4 \longrightarrow Na_2SO_4 + H_2O + CO_2}$ (1); $\mathrm{CO_2}$ turns lime water milky (1).	1+1+1
18	Baking soda: cakes, antacid (1); Washing soda: cleaning, softening hard water (1); Also used in glass and soap making (1).	1+1+1
	OR: Strong acid: red/orange (1); Neutral: green (1); Strong base: blue/violet (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Pass gas through lime water (1); Lime water turns milky (1); $\mathrm{Ca(OH)_2 + CO_2 \longrightarrow CaCO_3 + H_2O}$ (1); Milkiness disappears with excess $\mathrm{CO_2}$ (1); $\mathrm{CaCO_3 + H_2O + CO_2 \longrightarrow Ca(HCO_3)_2}$ (1).	1+1+1+1+1
	OR: Acid + base gives salt and water (1); $\mathrm{HCl + NaOH \longrightarrow NaCl + H_2O}$ (1); $\mathrm{H_2SO_4 + 2\,KOH \longrightarrow K_2SO_4 + 2\,H_2O}$ (1); Antacid for indigestion (1); Baking soda on bee sting (1).	1+1+1+1+1
20	Strong acid: pH 0-2, red; e.g. HCl (1); Weak acid: 3-6, orange/yellow; e.g. vinegar (1); Neutral 7, green; pure water (1); Weak base 8-10, blue; baking soda solution (1); Strong base 11-14, violet; NaOH (1).	1+1+1+1+1
	OR: Bacteria break down sugar to acid (1); pH falls below 5.5 (1); Enamel is corroded (1); Brush after eating to remove sugar (1); Basic toothpaste neutralises the acid (1).	1+1+1+1+1

Mistakes to avoid:

- Pouring water into concentrated acid; always add the acid slowly to water with stirring because the process is highly exothermic.
- Writing that a solution with pH 3 is 'less acidic' than one with pH 5; a lower pH means a **stronger** acid.
- Mixing up the formulae: baking soda is $\mathrm{NaHCO_3}$, washing soda is $\mathrm{Na_2CO_3 \cdot 10\,H_2O}$, plaster of Paris is $\mathrm{CaSO_4 \cdot \frac{1}{2}\,H_2O}$ and bleaching powder is $\mathrm{CaOCl_2}$.
- Thinking that all compounds containing hydrogen are acids; glucose and alcohol do not release $\mathrm{H^+}$ ions.
- Writing $\mathrm{H_2}$ as the gas formed when an acid reacts with a carbonate; the gas is $\mathrm{CO_2}$.

- Assuming neutralisation always gives a neutral solution; a strong acid with a weak base gives an acidic salt solution (and vice-versa).

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 2: **Acids, Bases and Salts** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** antacid	Antacids are mild bases.
2	**(d)** oxalic acid	Tomato contains oxalic acid.
3	**(b)** $NaHCO_3$	Sodium hydrogen carbonate.
4	**(c)** softening hard water	$Na_2CO_3 \cdot 10\,H_2O$ softens water.
5	**(d)** acidic solution	Pink only in base.
6	**(a)** $CaSO_4 \cdot \frac{1}{2}\,H_2O$	Hemihydrate of calcium sulphate.
7	**(d)** supporting fractured bones	Sets hard on mixing with water.
8	**(c)** alkalis	Alkalis are water-soluble bases.
9	**(c)** $CaSO_4 \cdot 2\,H_2O$	Dihydrate of calcium sulphate.
10	**(b)** disinfecting drinking water	Releases chlorine.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Lactic acid (1); Tartaric acid (1).	1+1
12	Litmus (1); Turmeric (also red cabbage, china rose) (1).	1+1
13	Neutral 7 (1); Acidic below 7, basic above 7 (1).	1+1
14	Sodium hydrogen carbonate (1); Sodium carbonate decahydrate (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Calcium chloride (1); Carbon dioxide (1); $CaCO_3 + 2\,HCl \longrightarrow CaCl_2 + H_2O + CO_2$ (1).	1+1+1
16	Bleaching cotton and linen (1); Disinfecting drinking water (1); Oxidising agent in industries (1).	1+1+1
17	CaO (1); $Ca(OH)_2$ (1); $CaCO_3$ (1).	1+1+1
	OR: Ionic compound from the reaction of an acid and a base (1); NaCl (1); Na_2SO_4, $CaCl_2$ (1).	1+1+1
18	Plaster for fractured bones (1); Making toys and statues (1); Smoothening walls (1).	1+1+1
	OR: Plaster of Paris and water (1); $CaSO_4 \cdot 2\,H_2O \longrightarrow CaSO_4 \cdot \frac{1}{2}\,H_2O + 1.5\,H_2O$ (1); Temperature should not exceed 373 K (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	NaOH: soaps, paper (1); Cl_2: water treatment (1); H_2: fuel, margarine (1); Process: electrolysis of brine (1); $2\,NaCl + 2\,H_2O \longrightarrow 2\,NaOH + Cl_2 + H_2$ (1).	1+1+1+1+1
	OR: Blue litmus: red in acid (1); Red litmus: blue in base (1); Phenolphthalein: colourless then pink (1); Methyl orange: red then yellow (1); Turmeric: stays yellow in acid, red in base (1).	1+1+1+1+1
20	$Ca(OH)_2 + Cl_2 \longrightarrow CaOCl_2 + H_2O$ (1); Bleaching powder: bleaching, disinfection (1); $NaCl + H_2O + CO_2 + NH_3 \longrightarrow NH_4Cl + NaHCO_3$ (1); Baking soda: baking, antacid (1); Formula $NaHCO_3$ (1).	1+1+1+1+1
	OR: $HCl + H_2O \longrightarrow H_3O^+ + Cl^-$ (1); Ions carry current (1); Bulb glows with HCl solution (1); Glucose does not form ions (1); Bulb does not glow with glucose (1).	1+1+1+1+1

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